Excelerate

Data Visualization Associate Early Internship

SLU 0408 DVA Team 40

4th Week Deliverables Report

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Week 4 – Compilation of Findings & Final Presentation (Team Report)

Internship: Data Visualization Associate (Excelerate Virtual Internship)
Week: 4 – Dashboard Refinement, Insights Extraction & Final Presentation
Team Contribution: Joint report covering dashboard analysis, key insights, and preparation of the final presentation for senior management.
Dashboard Report Link to View Access in Looker Studio :
https://lookerstudio.google.com/reporting/f5cee034-e374-4f6f-b313-264f8b375e4d

Introduction

Following the successful completion of Week 3, where the team developed a structured mapping table, identified key KPIs, and designed a detailed wireframe to guide dashboard creation, Week 4 focused on refining the fully developed dashboard and preparing for the final presentation. The emphasis shifted from building to interpreting, with the goal of translating raw data into actionable insights.

This stage required not only ensuring that the dashboard was visually clear, interactive, and business-ready but also demonstrating our ability to communicate findings effectively. By analyzing the dashboard outputs, the team compiled insights, identified trends, and extracted meaningful patterns that highlight the value of the selected KPIs. These findings were then structured into a professional presentation tailored for senior management.

Objectives

- **Analyze Dashboard Outputs:** Extract trends, correlations, and insights from key metrics.
- **Structure Professional Presentation:** Clearly communicate the methodology, process, and impact of findings.
- **Develop a Slide Deck:** Create a visually compelling and data-driven presentation highlighting KPIs, trends, and business implications.
- **Curate Final Dashboard:** Refine and optimize for clarity, usability, and impactful storytelling.

<u>Activities of Week-4</u>

1. **Insight Extraction:**
- Reviewing KPIs (applications, enrollment trends, learner engagement, demographic distributions).
- Identifying key trends and anomalies across opportunities, cohorts, and user demographics.
- Compiling actionable insights to support data-driven decision-making.

2. **Dashboard Refinement:**
- Ensuring filters, drill-downs, and chart placements enhanced user experience.
- Validating consistency of mapping table logic and business rules.
- Adjusting visualization formats to emphasize critical KPIs.

3. **Presentation Development:**
- Designing a slide deck highlighting methodology, data pipeline (Weeks 1–3), and (week-4) dashboard outputs.
- Structuring findings into context → insight → business implication flow.
- Incorporating visuals from the dashboard to strengthen storytelling.

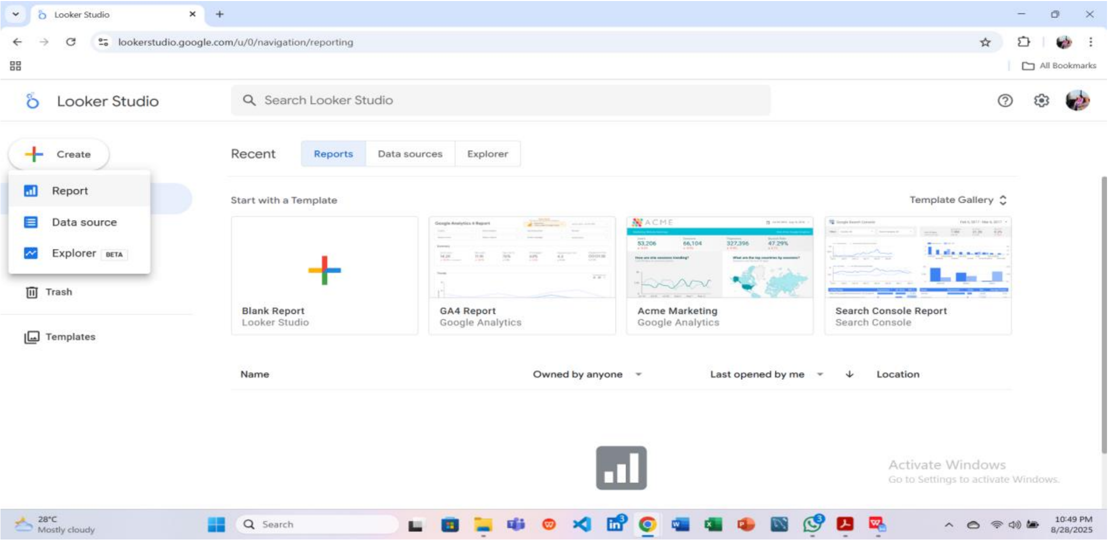
Expected Deliverables by the End of Week 4

- **Final Dashboard Link:** Refined and optimized for clear insights and interactivity.
- **Comprehensive Report:** Summarizing activities of Week-4, documentation of each details.

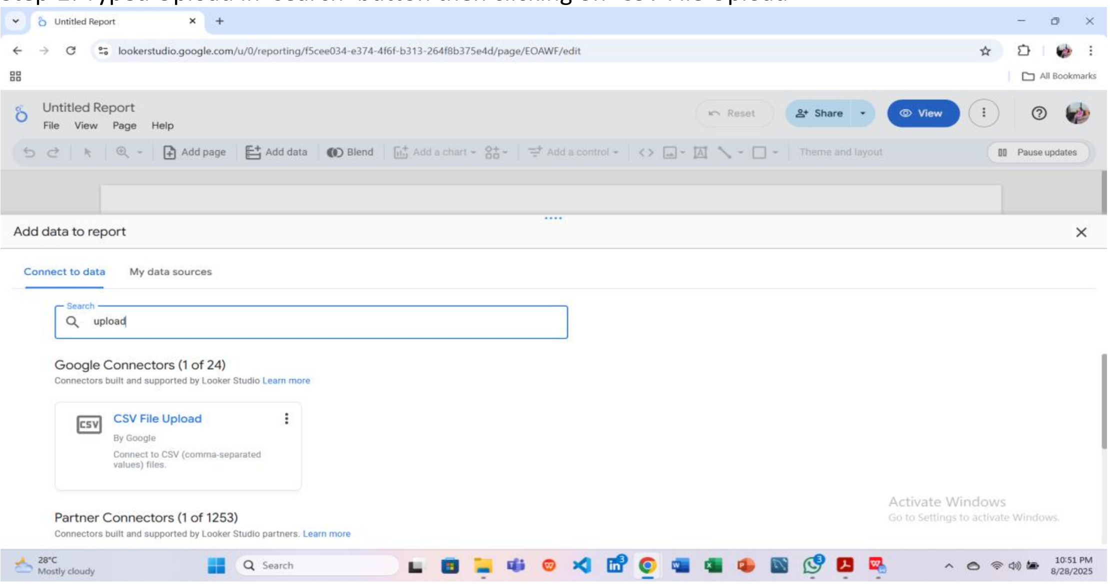
❖ **Team Presentation :** Professional slides highlighting of all week's activities including KPIs, insights, and business implications for senior management.Delivery of data-driven findings, showcasing the ability to interpret and communicate insights effectively.

Connection To Looker Studio:

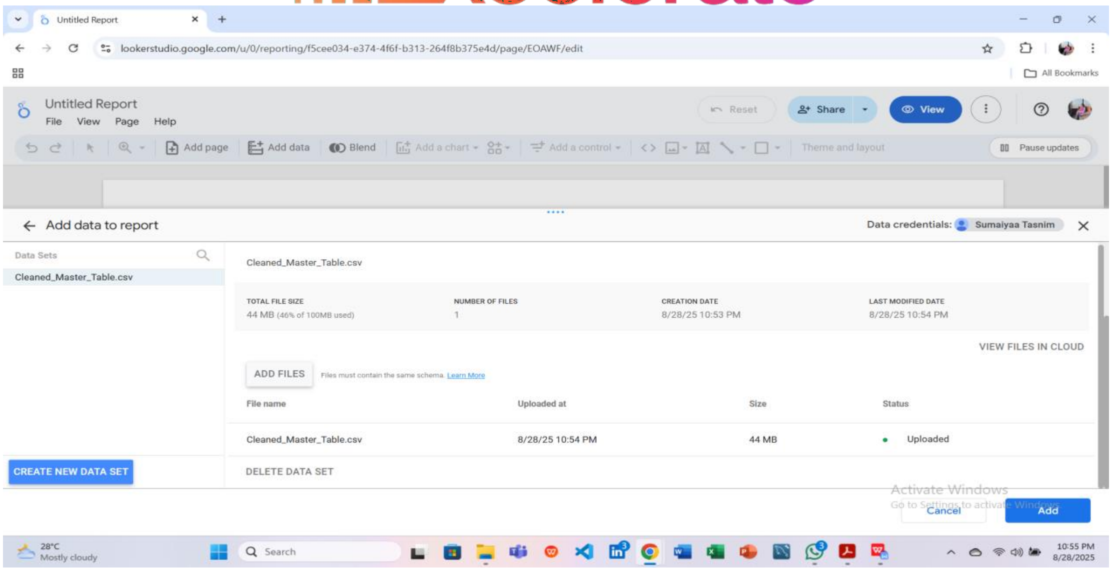
Step-1: clicking on 'Create' button & then clicking on 'Report'



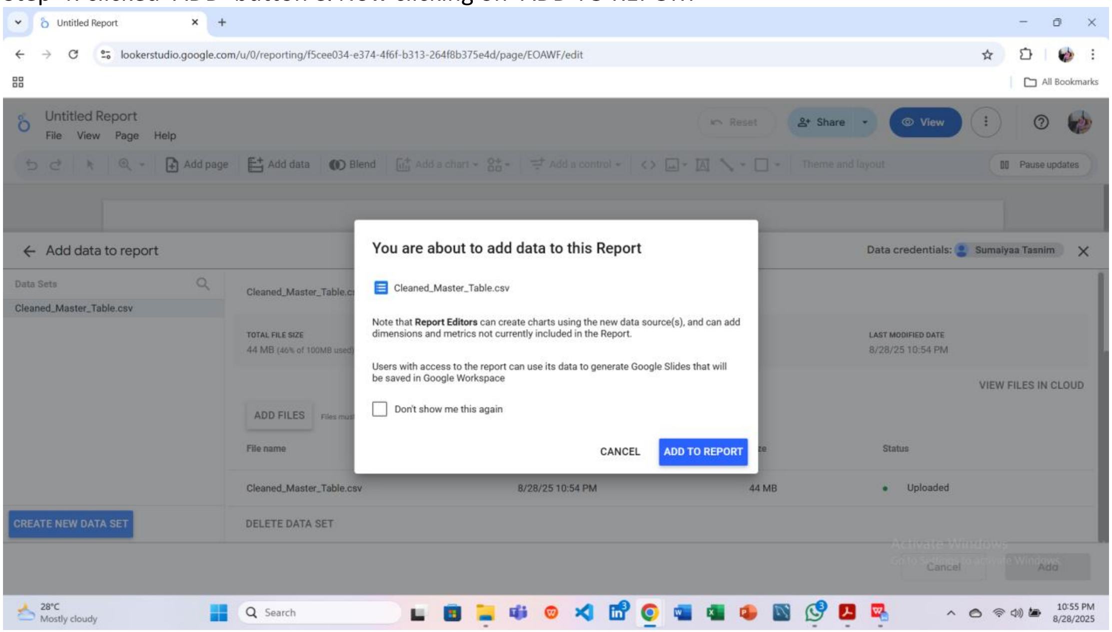
Step-2: Typed Upload in 'search' button then clicking on 'CSV File Upload'



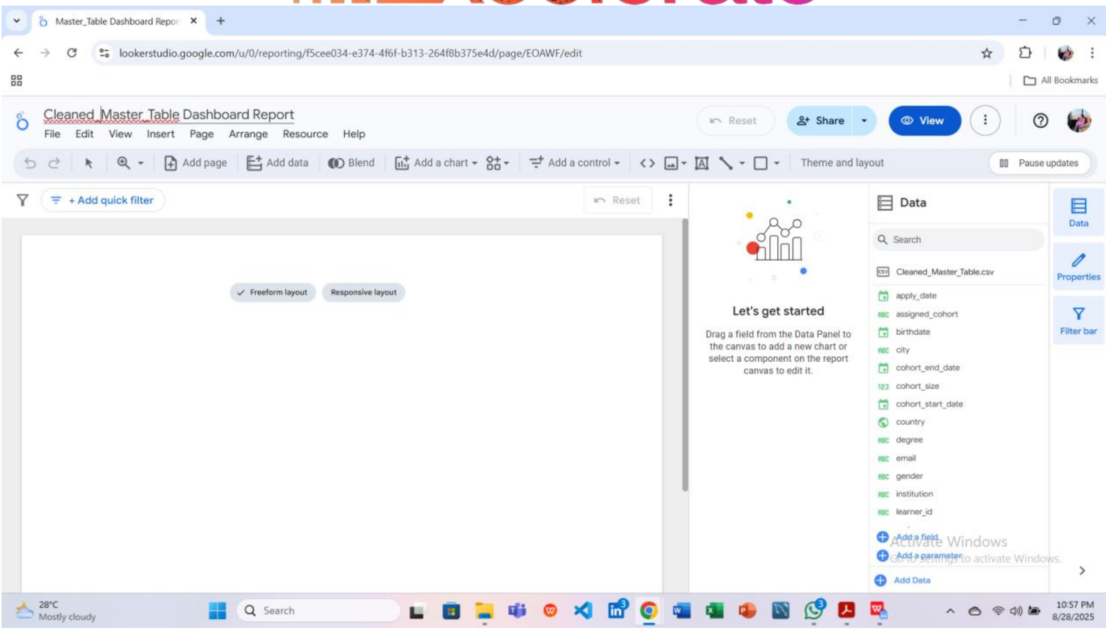
Step-3: Uploaded "Cleaned_Master_Table.csv" file



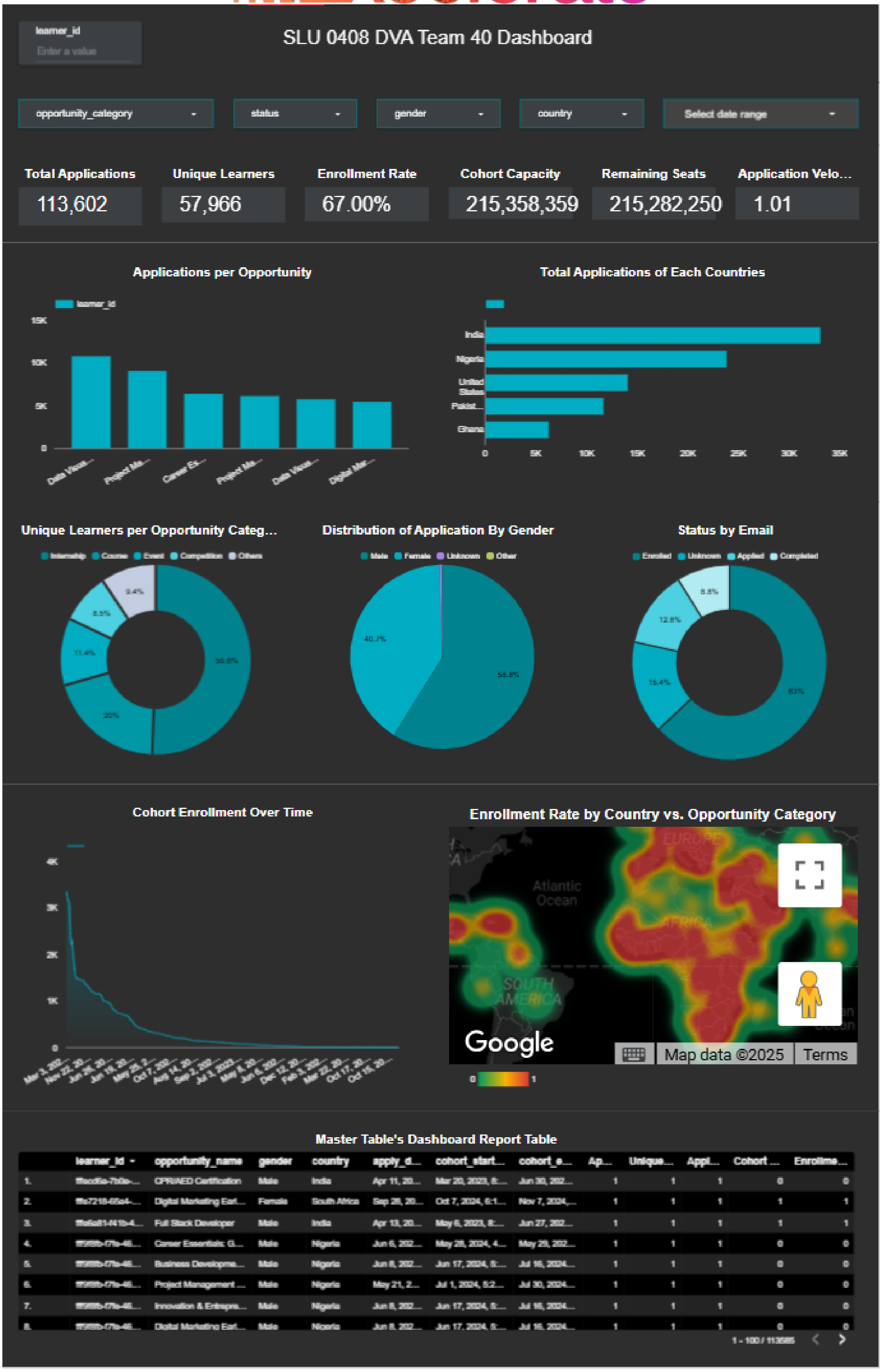
Step-4: clicked ‘ADD’ button & Now clicking on ‘ADD TO REPORT’



Step-5: Finally Imported Dataset of Cleaned_Master_Table & Connected to Looker studio



Dashboard of Master_Table:



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Details of Key Performance Indicators (KPIs) & Selected Relevent Metrics: (Followed Mapping Table)

KPI Name	Dimension (Formula)	Why Added
Total Applications	COUNT(*)	To measure overall application volume across all opportunities.
Unique Learners	COUNT_DISTINCT(learner_id)	To track how many individual learners are applying, avoiding duplicates.
Enrollment Rate	SUM(CASE WHEN status = 'Enrolled' THEN 1 ELSE 0 END) / COUNT(apply_date)	To evaluate the effectiveness of applications converting into enrollments.
Cohort Capacity	SUM(cohort_size)	To understand the total seats available for each cohort.
Remaining Seats	SUM(cohort_size) - SUM(CASE WHEN status = 'Enrolled' THEN 1 ELSE 0 END)	To monitor available capacity and plan for potential overflows.
Application Velocity	COUNT(*) / COUNT_DISTINCT(apply_date)	To measure the speed at which applications are coming in, useful for trend analysis.

Implementation Of Filters, Drill-Downs & Interactive Elements with Importance:

(Filters)

Filter	Suitable Filter Type	Notes / Usage
Opportunity Category	Dropdown / Multi-select	Allows filtering charts and KPIs by one or multiple opportunity categories.
Status	Dropdown / Single-select	Quickly isolate learners by their application status (Enrolled, Applied, etc.).
Gender	Dropdown / Single-select	Analyze demographic splits in charts like pie or bar charts.
Country	Dropdown / Multi-select	Filter data geographically for applications, enrollments, and enrollment rate.
Date Range (cohort_start_date → cohort_end_date)	Date Range / Slider	Analyze trends over specific cohorts, enrollment rate, and application velocity.
learner_id	Search Control (Equals)	Lookup a specific learner by ID → useful for user-level drill-down or debugging.

(Drill-Down Controls)

Dimension / Metric	Formula	User Interaction Features
learner_id (Dimension)	—	Can be used for drill-through to learner-level detail, search, or filtering.
opportunity_name (Dimension)	—	Supports drill-down (Opportunity Category → Opportunity Name), and filtering by program.
gender (Dimension)	—	Enables demographic filter or grouping (e.g., Male/Female/Other split).
country (Dimension)	—	Allows geographic drill-down and cross-filtering (row chart, heatmap).
apply_date (Dimension)	—	Used for time-based filtering, sorting, and trend exploration.
cohort_start_date (Dimension)	—	Allows filtering or grouping by cohort start → trend analysis over time.
cohort_end_date (Dimension)	—	Enables filtering/grouping by cohort completion → useful for capacity planning.
Applied (Metric)	COUNT(learner_id)	Allows sorting by application volume, comparison across dimensions.
Unique Learners (Metric)	COUNT_DISTINCT(learner_id)	Ensures drill-down to unique learners, prevents duplicate counts in interactions.
Applies per Country (Metric)	COUNT(apply_date)	Supports cross-filtering with Country dimension → easy country-level comparisons.
Cohort Enrollment (Metric)	SUM(CASE WHEN status = 'Enrolled' THEN 1 ELSE 0 END)	Enables trend and cohort performance analysis when combined with cohort dimensions.
Enrollment Rate (Metric)	SUM(CASE WHEN status='Enrolled' THEN 1 ELSE 0 END) / COUNT(apply_date)	Can be highlighted with conditional formatting → helps spot high/low conversion regions.

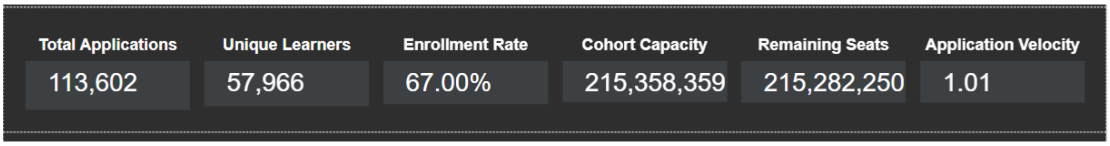
Every Chart's Details & Importance:

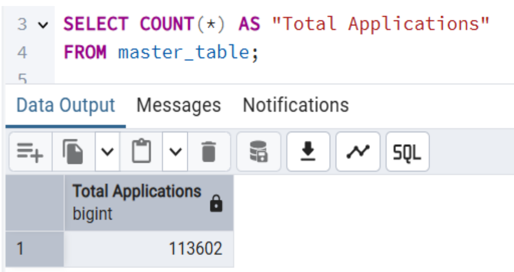
Chart Type	Chart Name	Dimension(s)	Metric (with Formula)	Reason & Benefit
Bar Chart	Applications per Opportunity	opportunity_name	COUNT(learner_id)	Shows how many learners applied to each opportunity → identifies program demand.
Row Chart	Total Applications of Each Country	country	COUNT(apply_date)	Compares total applications across countries → useful for geographic insights.
Donut Chart	Unique Learners per Opportunity Category	opportunity_category	COUNT_DISTINCT(learner_id)	Highlights learner diversity across categories → avoids duplicate counts.
Pie Chart	Distribution of Application By Gender	gender	COUNT(apply_date)	Shows gender split of applicants → useful for demographic and diversity analysis.
Donut Chart	Status by Email	status	COUNT_DISTINCT(email)	Displays learner distribution by application status → helps track pipeline stage.
Line Chart	Cohort Enrollment Over Time	cohort_start_date	SUM(CASE WHEN status = 'Enrolled' THEN 1 ELSE 0 END)	Tracks enrollment trends over cohorts → helps measure growth & conversion trends.
Heatmap	Enrollment Rate by Country vs. Opportunity Category	country	SUM(CASE WHEN status='Enrolled' THEN 1 ELSE 0 END) / COUNT(apply_date)	Reveals efficiency of enrollment across geography & categories → spot strengths & gaps.

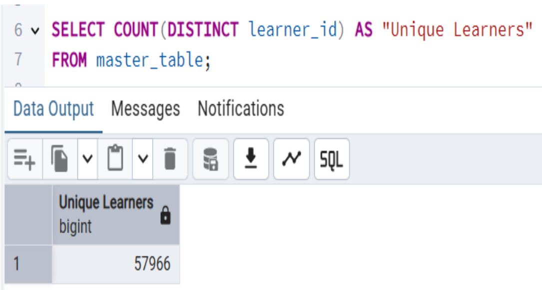
Review & Refining Visualizations:

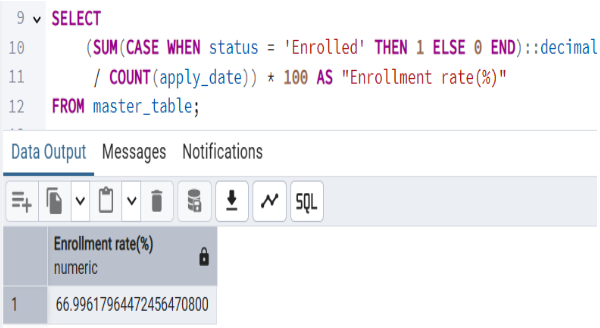
Comparison of dashboard's each value with the Output Values in PostgreSQL:

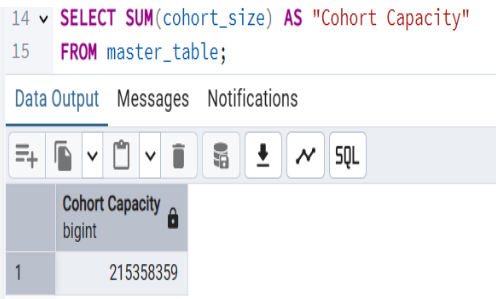
KPI(s) detection:

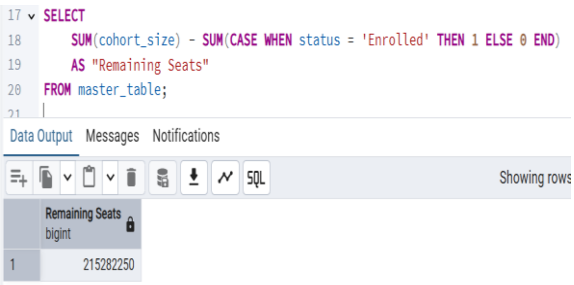


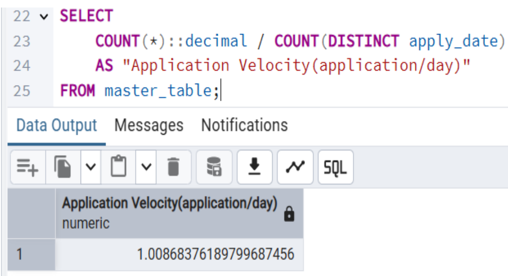












The output values obtained from PostgreSQL queries perfectly match the KPI values displayed on the dashboard.

Chart(s) Detection:

Bar Chart Values (Applications per Opportunity):

```sql
SELECT
    opportunity_name,
    COUNT(learner_id) AS "Total Applications"
FROM master_table
GROUP BY opportunity_name
ORDER BY "Total Applications" DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 170

	opportunity_name character varying (255)	Total Applications bigint
1	Data Visualization Early Internship	10772
2	Project Management Early Internship	9026
3	Career Essentials: Getting Started with Your Professional Journey	6345
4	Project Management Associate Early Internship	6102
5	Data Visualization Associate Early Internship	5704
6	Digital Marketing Early Internship	5435
7	CPR/AED Certification	5268
8	Jump Start: Developing your Emotional Intelligence	4005
9	Business Consulting Early Internship	3088
10	Innovation & Entrepreneurship Early Internship	2924
11	Digital Strategy Early Internship	2761
12	Linked Up: The LinkedIn Makeover Workshop	2430
13	Business Development Virtual Internship	2156

Total rows: 170 Query complete 00:00:00.171

Row Chart Values (Total Applications of Each Country) & Donut Chart Values (Unique Learners per Opportunity Category) :

```sql
SELECT
    country,
    COUNT(apply_date) AS "Total Applications"
FROM master_table
GROUP BY country
ORDER BY "Total Applications" DESC;
```

Data Output | Messages | Notifications

	country text	Total Applications bigint
1	India	33116
2	Nigeria	23887
3	United States	14125
4	Pakistan	11725
5	Ghana	6306
6	Kenya	5142
7	Egypt	3851
8	Philippines	3606
9	South Africa	3148
10	Bangladesh	2891
11	Nepal	1250
12	Vietnam	331
13	United Kingdom	269
14	Unknown	223

Total rows: 149 Query complete 00:00:00.256

```sql
SELECT
    opportunity_category,
    COUNT(DISTINCT learner_id) AS "Unique Learners"
FROM master_table
GROUP BY opportunity_category
ORDER BY "Unique Learners" DESC;
```

Data Output | Messages | Notifications

	opportunity_category character varying (100)	Unique Learners bigint
1	Internship	40240
2	Course	15933
3	Event	9074
4	Competition	6769
5	Masterclass	3998
6	Career	3366
7	Engagement	147

Pie Chart Values (Distribution of Application By Gender) & Donut Chart Values (Status by Email):

```sql
SELECT
    gender,
    COUNT(apply_date) AS "Total Applications",
    ROUND(
        COUNT(apply_date)::decimal * 100 / SUM(COUNT(apply_date)) OVER (),
        2
    ) AS "Percentage(%)"
FROM master_table
GROUP BY gender
ORDER BY "Total Applications" DESC;
```

Data Output Messages Notifications

	gender text	Total Applications bigint	Percentage(%) numeric
1	Male	66539	58.57
2	Female	46007	40.50
3	Unknown	966	0.85
4	Other	90	0.08

```sql
SELECT
    status,
    COUNT(DISTINCT email) AS "Unique Emails",
    ROUND(
        COUNT(DISTINCT email)::decimal * 100 / SUM(COUNT(DISTINCT email)) OVER (),
        2
    ) AS "Percentage(%)"
FROM master_table
GROUP BY status
ORDER BY "Unique Emails" DESC;
```

Data Output Messages Notifications

Showing rows: 1 to 4

	status text	Unique Emails bigint	Percentage(%) numeric
1	Enrolled	44995	62.96
2	Unknown	11028	15.43
3	Applied	9148	12.80
4	Completed	6298	8.81

Line Chart Values (Cohort Enrollment Over Time) & Heatmap Chart Values (Enrollment Rate by Country vs. Opportunity Category):

```sql
SELECT
    cohort_start_date,
    SUM(CASE WHEN status = 'Enrolled' THEN 1 ELSE 0 END) AS "Enrolled Count"
FROM master_table
GROUP BY cohort_start_date
ORDER BY cohort_start_date;
```

Data Output Messages Notifications

	cohort_start_date timestamp with time zone	Enrolled Count bigint
1	2022-06-09 13:33:20+06	1
2	2022-06-13 12:00:00+06	0
3	2022-07-01 10:33:20+06	0
4	2022-07-15 19:00:00+06	0
5	2022-07-15 21:46:40+06	0
6	2022-08-29 11:13:20+06	390
7	2022-09-02 09:40:00+06	112
8	2022-09-09 05:33:20+06	0
9	2022-09-16 07:00:00+06	0
10	2022-09-23 05:40:00+06	0
11	2022-10-10 08:46:40+06	0
12	2022-10-15 19:20:00+06	0
13	2022-10-17 10:13:20+06	0
14	2022-10-23 21:46:40+06	0

Total rows: 232 Query complete 00:00:00.200

```sql
SELECT
    country,
    opportunity_category,
    ROUND(
        SUM(CASE WHEN status = 'Enrolled' THEN 1 ELSE 0 END)::decimal
        / COUNT(apply_date) * 100, 2
    ) AS "Enrollment Rate(%)"
FROM master_table
GROUP BY country, opportunity_category
ORDER BY country, opportunity_category;
```

Data Output Messages Notifications

	country text	opportunity_category character varying (100)	Enrollment Rate(%) numeric
1	Afghanistan	Career	33.33
2	Afghanistan	Competition	87.50
3	Afghanistan	Course	100.00
4	Afghanistan	Event	80.00
5	Afghanistan	Internship	60.87
6	Afghanistan	Masterclass	100.00
7	Albania	Internship	66.67
8	Algeria	Career	100.00
9	Algeria	Competition	100.00
10	Algeria	Course	50.00
11	Algeria	Event	33.33

Total rows: 551 Query complete 00:00:00.241

When hovering over any chart in the dashboard, the corresponding values are displayed. These values have been validated by executing queries in PostgreSQL, and the outputs perfectly match the values shown on the dashboard.

Filters: All filters in the dashboard are functioning flawlessly, allowing users to refine data effectively by selecting or unselecting options.

Drill-Down Control: The drill-down functionality operates seamlessly, enabling users to explore data hierarchies and gain deeper insights with ease.

Layout, Colors, and Fonts: The dashboard's layout, color scheme, and font choices are visually appealing and well-structured, ensuring clarity and enhancing overall user experience.

The design maintains consistency and readability across all views, supporting intuitive navigation and interpretation of the data.

Data Accuracy: In PostgreSQL, all relevant queries were executed to validate the metrics, and the results obtained from the database perfectly matched the values displayed on the dashboard. No discrepancies or errors were observed. The dashboard accurately reflects the underlying data and is fully visual, ensuring reliable insights for users.

Dashboard Viewpoint: The dashboard is fully responsive and adapts seamlessly across different devices, including desktops, tablets, and mobile phones. All elements—charts, tables, filters, and text—fit perfectly within the frame, maintaining clarity and visual consistency. The layout, colors, and fonts remain well-structured, ensuring an optimal user experience on any screen size.

Conclusion

In Week 4, the team focused on turning the completed dashboard into meaningful insights and a clear presentation for senior management. The dashboard was refined to be easy to use, visually appealing, and fully interactive, with all filters and drill-downs working smoothly. It is also fully responsive, adapting seamlessly across desktops, tablets, and mobile devices, while maintaining clarity, visual consistency, and a well-structured layout, colors, and fonts.

By analyzing key metrics like applications, enrollment trends, learner demographics, and many more - the team uncovered important patterns and trends that can guide decision-making. All data was carefully checked in PostgreSQL to ensure it matched perfectly with what's shown on the dashboard, guaranteeing accuracy and reliability.

The final outputs—a polished dashboard, a detailed report, and a professional slide deck—showcase the team's ability to not just present data, but to tell a story with it. Overall, the project demonstrates how clear visualization and analysis can turn raw data into actionable insights.